

Experiment 9

Setting Up a Wireless Network

Lab Objective:

To **configure** and **verify** both **IPv4** and **IPv6** addressing on wireless network devices like routers, switches, access points, and client devices (PCs/laptops).

What is a Wireless Network?

A **wireless network** allows devices to stay connected to the network without physical cables. It uses **radio frequency (RF)** waves to connect devices like laptops, mobile phones, or desktops to the internet and internal networks.

Key components:

- **Wireless Router / Access Point (AP)**
- **Client Devices** (like laptops, smartphones)
- **SSID (Service Set Identifier)** – the network name
- **Security Protocols** – WPA2/WPA3

◆ What is IP Addressing?

IP addressing is used to **uniquely identify** devices on a network. It enables **communication** between devices.

IPv4 Address:

- 32-bit address, written in **dotted decimal format** (e.g., 192.168.1.1)
- Consists of **Network ID** and **Host ID**

- Divided into **Classes (A, B, C)** or **Private/Public** ranges

IPv6 Address:

- 128-bit address, written in **hexadecimal colon-separated** format (e.g., 2001:0db8:85a3::8a2e:0370:7334)
- Supports **vast number of addresses** (2^{128})
- Eliminates the need for NAT
- More secure and efficient

Steps to Configure IPv4 and IPv6 Addressing

1. Set Up Wireless Router / Access Point

- Access the router through its default IP (192.168.0.1 or 192.168.1.1)
- Set **SSID (network name)** and **password**
- Enable **DHCP** for automatic IPv4 assignment
- Configure **IPv6** if supported by the device (Static or SLAAC)

2. Configure IPv4 on Client Device

- Go to **Network Settings** on the PC/laptop
- Select the wireless network and connect using the password
- If using **manual IPv4 configuration**:
 - IP address: 192.168.1.10
 - Subnet mask: 255.255.255.0
 - Gateway: 192.168.1.1
 - DNS: 8.8.8.8

3. Configure IPv6 on Client Device

- If automatic, the system will get an address via SLAAC or DHCPv6
- If manual:
 - IPv6 address: 2001:db8:abcd:001::1
 - Prefix length: 64
 - Default gateway: 2001:db8:abcd:001::ff
 - DNS: 2001:4860:4860::8888

Verification Commands

On Windows Command Prompt:

ipconfig /all

- Shows IPv4 and IPv6 addresses assigned.

```
ping 192.168.1.1  
ping google.com  
ping -6 google.com
```

- Verifies connectivity for both IPv4 and IPv6

On Router (via CLI like Cisco IOS):

```
show ip interface brief  
show ipv6 interface brief
```

- Displays configured Ips

```
ping 192.168.1.10  
ping 2001:db8:abcd:001::1
```

Important Configuration Commands (Cisco Router Example)

```
enable  
configure terminal  
  
interface GigabitEthernet0/0  
  ip address 192.168.1.1 255.255.255.0  
  ipv6 address 2001:db8:abcd:1::1/64  
  no shutdown  
  
exit
```

Conclusion / Outcome:

By the end of this experiment, you should be able to:

- Set up a basic wireless network.
- Configure and assign both IPv4 and IPv6 addresses.
- Verify IP configurations using commands.
- Understand the differences between IPv4 and IPv6.
- Ensure network connectivity and proper address assignment.